XMBA 201A – Managerial Economics

Office Hours Practice Problems: Lectures 1–5 Review

GSI: Simoni Jain | Date: Oct 21, 2026

Question 1: Software Pricing

Based on Lecture 5

You have a patent on a unique software package with an application to business procedures. The version of the product your company developed is suited for medium to large business, but you could disable some of the features in order to create a less valuable "small business" version that is sufficient for small businesses. Since the work has been the development of the algorithms, both products effectively have zero marginal cost. Suppose you have two groups of potential customers. Thirty are medium to large business who are willing to pay up to $2M for the Full Version (F) of your product and $0 for the Small-Business Version (S). Seventy are small businesses are willing to pay up to $0.7M for F and $0.5 for S. Each business will choose the product that gives it the highest net surplus.

a) If you offer only the F version of your product, what price would you charge and what profits would you make?

b) If you offer both versions of the product, what prices would you charge and what profits would you make?

c) Now suppose that instead of valuing the S version at $0, the medium to large businesses value the S version at $1.2M, with all other values the same as before. If you stuck to the prices you found in part (b) above, what would your profits be? In this case, would you prefer to choose the simple price you charged in part a above?

d) Continue to assume that the medium to large businesses value the S version at $1.2M, with all other values the same as before. What prices would you charge if you sold both products? Is this better or worse than selling only the F version?

	S	F
Small (n=70)	$0.5 m	$0.7m ✗
Large (n=30)	$0 m	$2 m ✓

MC = 0

a) Only F version

$\pi(p=2m) = 30 \times 2 = 60$ m

$\pi(p=0.7m) = 100 \times 0.7 = 70$m ✓

Price F at 0.7m.

b) 30×2.0 ✓ $= 60$ m. for F-version ✓

$70 \times 0.5 = 35$ m for S-version ✓

$95 m ✓

c)

	S	F
small (n=70)	$0.5 m	$0.7m
Large (n=30)	1.2 m	$2 m

In part b, price F at $2m & S at $0.5m

Surplus (If Large buys S): $1.2 - 0.5 = \$0.7m$

" (If Large buy F): $2 - 2 = \$0m$

$100 \times 0.5 = \$50m$

d) Charge $0.5 m for S ✓

$(0.5 + 0.7 = 1.2\ S)$

The surplus that Large will get if they buy S: $1.2 - 0.5 = \$0.7m$

Price F s.t. $2 - p = 0.7$

$p = 2 - 0.7$

$= \$1.3m$

S at 0·5m & F at 1·3m

30 × 1·3 + 70 × 0·5 = 74 m

Question 2: Sequential Uncertainty and the Value of a Marketing Deal

Based on Lectures 1 and 2

Fall 2022 Midterm

Amazon is considering developing an augmented reality eyewear product, to succeed where Google Glass failed. This will require a two-phase process: the first phase is product development and the second phase is hiring a marketing company to get consumers excited about the product. Both phases must be successfully completed before Amazon can reap the benefits from the product. However, management is aware that there is a chance that one or both phases might fail, in which case the product will not sell. Product development must happen first, and if it fails there is no point in hiring the marketing company.

Product development costs $1 billion and has a 0.5 probability of success. Conditional on having a successful product, the marketing company will charge Amazon $500 million, and has a 0.8 probability of success. The risk of failure of the marketing campaign is uncorrelated with the risk of failure of product development. If both steps are successful, Amazon expects total net present value of profits from the product to be $4 billion. Amazon is, of course, risk neutral.

a) [10] Draw Amazon's complete decision tree.

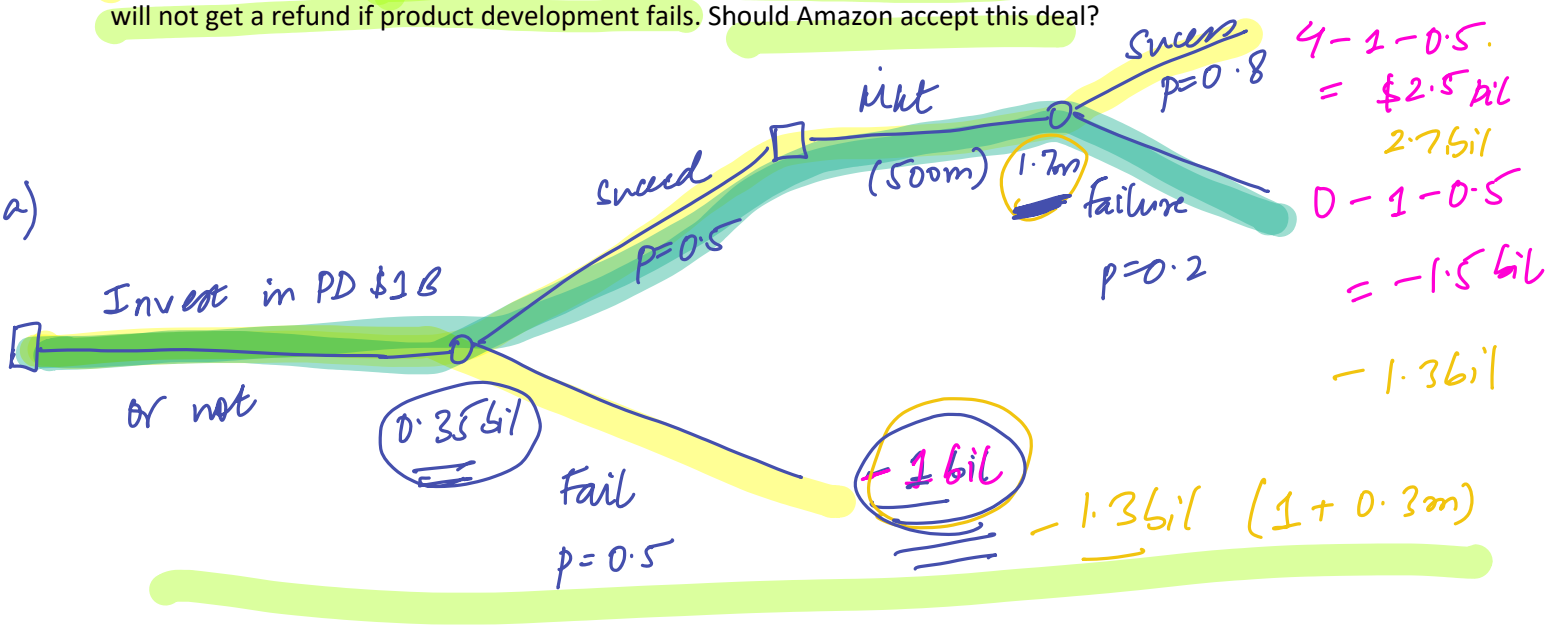
b) [5] What is the probability that Amazon succeeds if it chooses to pursue the product?

$0.5 \times 0.8 = 0.4$

c) [5] What is the expected value to Amazon from launching the product development phase?

d) [15] A competing marketing company that wants to get Amazon's business offers Amazon a sweet deal: they will provide a campaign at a cost of only $300 million, and they will deliver the same odds of success as the more expensive company. However, Amazon must pay them in advance of product development to retain them and will not get a refund if product development fails. Should Amazon accept this deal?

a)



c) $0.8 \times 2.5 + 0.2(-1.5) = 2 - 0.3 = \$1.7m$

$0.5 \times 1.7 + 0.5(-1) = 0.85 - 0.5$

$= \$0.35m$

d) $EY(\text{success}) = 0.8 \times 2.7 + 0.2(-1.3) = 1.96$ bil

$EY(\text{deal}) = 0.5 \times 1.9 + 0.5(-1.3)$

$= 0.95 - 0.65$

$= \boxed{\$0.306}$

Question 3: Pricing the Kindle

Based on Lecture 4

When Amazon introduced the Kindle in 2007, its price was an impressive $359. In 2009 Amazon dropped the price to $259, and in 2010 to $189. Financial analysts estimated annual sales of about 1 million units at the $259 price point, and that sales roughly tripled to about 3 million units after the cut to $189. These two data points imply a linear demand curve of P = $295 − 35Q, where Q is in millions of units per year. The estimated marginal cost of producing a Kindle is $129.

a) [5] Using MR = MC, find the profit-maximizing quantity and price for the Kindle, ignoring any revenue beyond the device itself.

b) [10] Amazon earns a profit contribution of about $4 per e-book sold, and each Kindle sold generates on average 25 e-book sales — an additional $100 of expected profit per device sold. Taking this into account, find the new profit-maximizing quantity and price.

c) [5] Does this help explain Amazon's decision to cut the Kindle's price from $259 to $189? Explain.

		Qty D
In 2007	P = $259	
2009	P = $259	1 million
2010	P = $189	3 million

a) $P = 295 - 35Q$

$TR = P.Q = 295Q - 35Q^2$

$MR = \frac{\partial TR}{\partial Q} = 295 - 35.2Q$

$P = a - bQ$

$MR = a - 2bQ$

$295 - 70Q = 129$

$\Rightarrow Q^* = \frac{295 - 129}{70} = \frac{166}{70} = 2.37$ m units.

$P^* = 295 - 35 \times \frac{166}{70} = 83$

$= \$212$

b) $MR(Q) + 100 = MC$

$295 - 70Q + 100 = 129$

$\frac{395 - 129}{70} = Q^*$

$\frac{266}{70} = Q^*$

$3.8\ m = Q^*$

$P^* = 295 - 35(3.8)$

$= \$162$